

Rethinking Decoder Redundancy in Efficient 3D Medical Image Segmentation: A Systematic Characterization Study

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Abstract

001 *The decoder is an essential module for fusing multi-scale*
002 *features and restoring spatial detail in U-Net architectures*
003 *for 3D medical image segmentation. Recent designs, espe-*
004 *cially attention-based decoders, sharply increase its cost,*
005 *and efficient methods such as EffiDec3D respond by shrink-*
006 *ing high-resolution decoder stages—removing > 85% of its*
007 *parameters and FLOPs across 3D UX-Net, MedNeXt-M-*
008 *K3 and SwinUNETR with little loss in Dice. This success,*
009 *however, rests on the empirical assumption that decoder*
010 *computation is static and globally redundant; it does not*
011 *explain why the computation is removable, where the de-*
012 *coder still matters, or whether those regions can be identi-*
013 *fied at test time. We present a systematic characterization of*
014 *decoder redundancy along three axes—heterogeneity, con-*
015 *centration, and predictability—by counting per-voxel pre-*
016 *diction flips between matched full and efficient decoders,*
017 *with a frozen shared-encoder control that attributes the*
018 *changes to the decoder alone and a channel/resolution fac-*
019 *tor decomposition that separates the two sources of re-*
020 *dundancy. We find the full decoder’s net benefit is small*
021 *and bidirectional (about 0.1% of voxels), spatially concen-*
022 *trated at organ boundaries and small structures, and identi-*
023 *fiable from the efficient model’s own uncertainty—entropy-*
024 *ranked regions recover its net benefit at roughly five times*
025 *the matched-random rate at a fixed budget. These trends*
026 *are consistent across three architectures and three datasets,*
027 *indicating that decoder redundancy is structured and pre-*
028 *dictable rather than uniform. Our analysis provides an*
029 *evaluation protocol and design guidance for more efficient,*
030 *region-adaptive segmentation decoders.*

031 1. Experimental Setup

032 **Experiment plan.** Our evaluation study reproduces Ef-
033 fiDec3D [11] and evaluates it with an *L-shaped* proto-
034 col (Table 1) that separately tests cross-dataset and cross-
035 architecture consistency without an expensive full facto-

Table 1. L-shaped protocol. The 3D UX-Net row tests cross-dataset consistency; the BTCV column tests cross-architecture consistency; they meet at the UX-Net/BTCV anchor (*, completed). Off-axis cells (–) are not run.

Backbone	BTCV (CT)	FeTA (MRI)	Hepatic Vessel (CT)
3D UX-Net	✓*	✓	✓
Swin UNETR	✓	–	–
MedNeXt-M-K3	✓	–	–

036 rial. A **dataset axis** runs the matched 3D UX-Net [8]
037 pair on three diverse tasks from the EffiDec3D evaluation—
038 BTCV abdominal CT, FeTA [10] fetal-brain MRI, and MSD
039 Task08 HepaticVessel [1] (CT, thin vessels/small lesions)—
040 and an **architecture axis** runs matched pairs on BTCV
041 for three backbone families: 3D UX-Net (large-kernel
042 CNN), Swin UNETR [13] (hierarchical Transformer), and
043 MedNeXt-M-K3 [12] (ConvNeXt-style CNN). The two
044 axes share the 3D UX-Net/BTCV anchor, which also car-
045 ries the mechanism analysis below.

046 **Dataset and split.** The primary task is the 13-organ
047 BTCV/Synapse abdominal CT benchmark with 18 training
048 and 12 validation volumes: spleen, right and left kidney,
049 gallbladder, esophagus, liver, stomach, aorta, inferior vena
050 cava, portal and splenic veins, pancreas, and right and left
051 adrenal glands. Images are intensity-windowed and resam-
052 pled following the public EffiDec3D pipeline; the same val-
053 idation subjects are paired across all model comparisons on
054 a given dataset.

055 **Two comparison regimes.** Within each cell the full and
056 EffiDec3D models share the same encoder *architecture* and
057 differ in the decoder (channels reduced to the minimum
058 encoder width, additive skip aggregation, resolution factor
059 two). We analyze them under two regimes. (i) **End-to-**
060 **end (ecological).** Both models are trained independently,

so they share the encoder architecture but not its weights; their flips reflect the deployed full-vs-efficient difference but conflate decoder capacity with encoder-weight and optimization differences. (ii) **Frozen shared encoder (causal).** We take one trained encoder, freeze it, and train the decoder variants on *identical* features, so prediction differences are attributable to the decoder alone. Both regimes hold the encoder architecture fixed, so the study characterizes the decoder and draws no conclusions about the encoder itself.

Frozen-encoder factor decomposition. EffiDec3D removes decoder computation in two ways at once—narrower channels and a dropped high-resolution stage. To attribute effects, we decompose them with a 2×2 factorial on the frozen encoder, all within the EffiDec3D decoder family so only the two knobs vary: full (highest-resolution stage, wide channels), channel-only (wide $\rightarrow$ 48 channels), resolution-only (drop the high-resolution stage), and the combined EffiDec3D decoder. Pairwise flips (Full $\rightarrow$ channel-only, Full $\rightarrow$ resolution-only, Full $\rightarrow$ combined) isolate which factor changes which regions. The frozen encoder is optimized for the full decoder, so “efficient still matches on frozen features” is a *conservative* redundancy result.

Null-pair seed control. Because the net flip is tiny, we bound seed noise with a same-architecture null pair: two Full models trained with different (predeclared) seeds. The claim is that Full-vs-EffiDec3D net flips and their concentration exceed this Full-vs-Full null.

Implementation. We reproduce the EffiDec3D training protocol on a single NVIDIA RTX 5090 (32 GB, Blackwell) under PyTorch 2.6 / CUDA 12.8 with MONAI, using BF16 mixed precision (no loss scaling). Each model is trained for 45,000 optimizer steps (validation every 250) with AdamW at learning rate 10^{-3} , a Dice-cross-entropy loss, four 96^3 random-crop samples per volume, and full in-memory caching. Sliding-window inference uses a 96^3 window, batch size four, overlap 0.7, and constant blending. We evaluate one full run and two independently initialized EffiDec3D runs per cell to expose seed sensitivity. Segmentation quality is reported with Dice and 95th-percentile Hausdorff distance (HD95), and efficiency with parameters, multiply-accumulate operations (MACs), latency, and peak memory.

Core quantities. Let $\hat{y}_E(v)$, $\hat{y}_F(v)$, and $y(v)$ be the efficient prediction, full prediction, and ground-truth label at voxel v . We measure where the full decoder changes predictions with *prediction flips*, the standard tool for comparing

two classifiers on paired data [2, 14]:

$$P(v) = \mathbb{I}[\hat{y}_E(v) \neq y(v) \wedge \hat{y}_F(v) = y(v)], \quad (1)$$

$$N(v) = \mathbb{I}[\hat{y}_E(v) = y(v) \wedge \hat{y}_F(v) \neq y(v)], \quad (2)$$

$$U(v) = P(v) - N(v), \quad (3)$$

a *positive* flip (full corrects efficient), a *negative* flip (full degrades efficient), and the *net* flip. P and N are exactly the discordant cells of the paired-agreement table behind McNemar’s classifier-comparison test [3, 9]; volume rates are $R_{\text{pos}} = \text{mean}_v P(v)$, $R_{\text{neg}} = \text{mean}_v N(v)$, and $R_{\text{net}} = R_{\text{pos}} - R_{\text{neg}}$. As test-time signals computable from the efficient model alone we use predictive entropy $H(v) = -\sum_c p_c(v) \log p_c(v)$, confidence $\max_c p_c(v)$, and Monte-Carlo dropout variance [4].

Characterization metrics. We group the analysis into three questions; each metric is computed per grid cell and aggregated over subjects with a bootstrap, since the twelve validation volumes—not the correlated voxels—are the independent units.

(1) *Heterogeneity: is the benefit evenly distributed?*

- **O1, boundary error ratio.** Mean voxel error on the eroded boundary band divided by the eroded-interior error; >1 indicates boundary-concentrated error.
- **O4, per-organ difficulty.** Per-class Dice and mean entropy H , ranking the structures the decoder struggles on.
- **O10, size-difficulty relation.** Spearman ρ between organ volume and difficulty, and the partial correlation of H and error given log-volume.
- **Boundary-resolved flips.** Positive/negative/net flip rate binned by Euclidean distance to the nearest GT boundary—testing whether the decoder *benefit* (not just error) concentrates near boundaries. Complemented by Normalized Surface Dice (NSD) at 1–2 voxel tolerance for a surface-level view.

(2) *Concentration: does a small subset carry most net benefit?*

- **O2, entropy sparsity.** $\Pr[H(v) > \tau]$, the fraction of high-entropy voxels.
- **O5, uncertainty-benefit correlation.** Subject-level Pearson $r(H, U)$ between binned entropy and net flips.
- **O9^{orc}, voxel recovery.** $\sum_{\text{top-}k\%} P(v) / \sum P(v)$, the share of positive flips in the highest-entropy $k\%$ of voxels—a non-deployable upper bound.
- **O6, training persistence.** Mean entropy across checkpoints, testing whether concentration is a training transient.

(3) *Predictability: can a test-time signal find the regions?*

- **O3, flip discrimination and calibration.** AUROC and AUPRC of entropy predicting flips [7], and expected calibration error (ECE) [6].

Table 2. Reproduction fidelity of the matched 3D UX-Net pair, per dataset. We report mean Dice of our runs against the published EffiDec3D value; Δ is our Effi-UX minus the published Effi-UX. Data were reconstructed with identity affine metadata, so this is a controlled reimplement, not an exact reproduction.

Dataset	Full-UX	Effi-UX	pub. Effi-UX	Δ
BTCV (CT, 13-organ)	.792	.777	.793	-.016
FeTA (MRI)	—	—	—	—
HepaticVessel (CT)	—	—	—	—

[TODO: Fill FeTA/HepaticVessel rows as the dataset-axis cells complete.]

• **O9, opportunity curve.** A risk–coverage curve [5] over contiguous 16^3 blocks (plus a 4-voxel halo) ranked by entropy, reporting net-flip utility recovered versus a matched-random selector.

• **O11, routing-signal comparison.** Subject-level correlation of entropy, confidence, and MC-dropout with net flips.

Cross-dataset (O7) and cross-backbone (O8) consistency re-evaluate these metrics across the rows and columns of Table 1.

[TODO: Completed results below are the 3D UX-Net/BTCV grid cell. Remaining cells (other backbones and datasets) are in progress; cross-backbone (columns) and cross-dataset (rows) consistency are reported once enough cells are filled.]

2. Results

2.1. Controlled Reimplementation

We reproduce the matched Full-UX/Effi-UX pair and report its fidelity per dataset (Table 2). On BTCV our Effi-UX reaches 0.777 mean Dice, 1.5 points below the published 0.793, while Full-UX is within 0.6 points of its published value; with EffiDec3D’s default concatenation skip our Effi-UX matches its published 3.15 M parameter count, so the residual gap reflects our reconstructed identity-affine data rather than the paper’s processed set (a FP32-versus-BF16 check ruled out a numerical cause). Efficiency reproduces the published reductions: Effi-UX uses $11.7\times$ fewer MACs, $16.8\times$ fewer parameters, $\sim 5\times$ lower latency, and $\sim 7.7\times$ lower peak memory. The same pattern holds on the architecture axis (Table 3): on BTCV, Swin UNETR reproduces its published full model (0.805 vs. 0.8013) and its EffiDec3D decoder cuts MACs $6.5\times$ at a 2.6-point Dice cost—a smaller reduction than 3D UX-Net’s, matching EffiDec3D’s own per-backbone figures. MedNeXt-M-K3, the strongest backbone at 0.837 full Dice, shows a 2.3-point cost at a $2.2\times$ MAC reduction; across the three backbones the Dice cost is small and of the same order (1.5–2.6%) even as the achievable MAC reduction ranges from

Table 3. Architecture axis (BTCV): reproduction and efficiency of the matched full/EffiDec3D pair per backbone. Dice is Full $\rightarrow$ EffiDec3D; reductions are Full/EffiDec3D.

Backbone	Dice (F $\rightarrow$ E)	MAC $\downarrow$	Param $\downarrow$	Lat $\downarrow$
3D UX-Net	.792 $\rightarrow$.777	$11.7\times$	$16.8\times$	$5.1\times$
Swin UNETR	.805 $\rightarrow$.779	$6.5\times$	$5.7\times$	$2.3\times$
MedNeXt-M-K3	.837 $\rightarrow$.815	$2.2\times$	$3.0\times$	$2.2\times$

Table 4. Prediction-flip rates for the parameter-matched (concatenation) efficient decoder (fractions of evaluated voxels). Positive/negative flips are the discordant cells of the paired comparison [9, 14]. The net-flip CI includes zero.

Metric	value
Positive flip rate R_{pos}	.00278
Negative flip rate R_{neg}	.00232
Net flip rate R_{net}	.00047
Net flip rate, subject 95% CI	[-.00026, .00112]

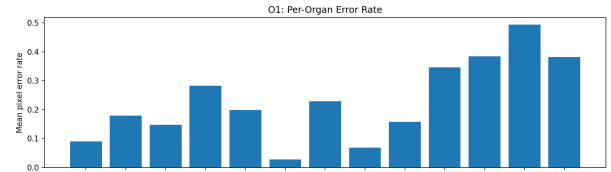


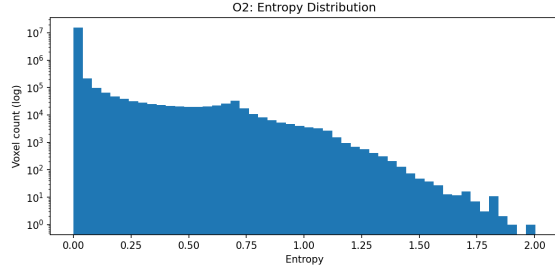
Figure 1. Per-organ error rate for Effi-UX (seed 1). Difficulty is strongly uneven across anatomy.

$2.2\times$ to $11.7\times$ —consistent across architectures, not universal in magnitude. Treating these matched models as fixed, the rest of the paper characterizes *where* the full decoder’s extra computation changes predictions—the question EffiDec3D’s aggregate evaluation leaves open.

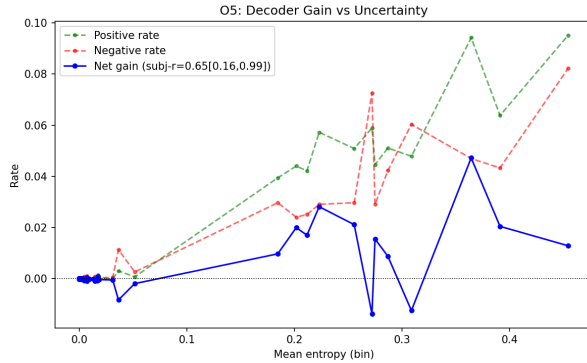
2.2. Heterogeneity Metrics

The full decoder’s benefit is small and *bidirectional*: Full-UX corrects 0.278% of Effi-UX’s voxels (R_{pos}) while degrading 0.232% (R_{neg}), so the net flip rate R_{net} is only 0.047% of voxels—and its subject-level 95% CI, $[-0.026\%, +0.112\%]$, *includes zero* (Table 4). With the parameter-matched Effi-UX, the full decoder’s net benefit is thus not even significantly positive: reporting positive flips alone would overstate a benefit that, netted against regressions, is statistically indistinguishable from none.

This small net benefit is far from uniformly distributed. As a first cut, Effi-UX error concentrates at structure boundaries ($4.39\times$ the eroded-interior rate) and on small, hard organs—the adrenals, pancreas, and veins—with net benefit rising as structure size falls (organ-size/net-flip $\rho = -0.37$; Fig. 1). Resolving the *net flip* rate by distance to the GT



(a) Voxel entropy (log-count).



(b) Net flips vs. entropy.

Figure 2. Concentration of difficulty and net flips (seed 1). Uncertainty is spatially sparse (a), and net flips rise with it (b).

boundary shows the decoder’s benefit is sharply boundary-localized: it peaks at the surface (0.085 at 0–1 vox) and is near zero or slightly negative beyond 1 vox; surface Dice (NSD) is 0.741/0.845 at $\tau=1/2$ mm. This concentration should exceed a same-architecture null pair (Full seed 0 vs. seed 1) [TODO: null-pair floor], ruling out seed noise.

2.3. Concentration Metrics

A small, high-uncertainty subset carries most of the net benefit. Predictive entropy is spatially sparse: the median voxel entropy is near zero and only 1.26% of voxels exceed 0.5 (Fig. 2a). What net flips there are concentrate in this tail—the subject-level Pearson correlation between binned entropy and net flips is $r = 0.639$ (95% CI [0.198, 0.987]), bounded above zero (Fig. 2b). A voxel-level oracle makes the concentration explicit: ranking foreground voxels by entropy, the top 20% contain 83.1% of all positive flips.

2.4. Predictability Metrics

Concentration is only actionable if the beneficial regions can be found without the ground truth. We assess this as an error-discrimination problem: predictive entropy should separate flip voxels from the rest [274]. Efficient-decoder entropy predicts the *positive-flip* voxels [275] with per-subject AUROC 0.903 (95% CI [0.876, 0.926]) and AUPRC 0.299 ([0.259, 0.342]); predicting the error voxels [276]

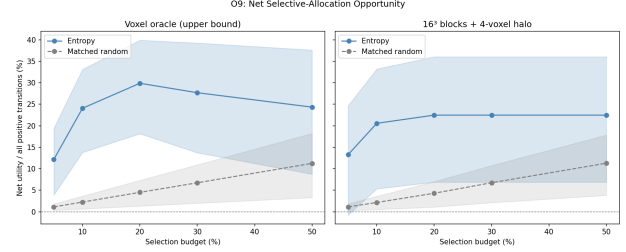


Figure 3. Region-level opportunity (seed 1). Contiguous 16^3 blocks ranked by entropy recover far more *net* flip utility than a matched-random selector at every budget $\geq 10\%$; the left panel is a non-deployable voxel oracle.

a cheaper baseline—gives AUROC 0.856 ([0.789, 0.897]) and AUPRC 0.530 ([0.499, 0.559]), and entropy is moderately calibrated (ECE 0.088, [0.065, 0.115]) [6]. The pooled-bin entropy-error correlation $r = 0.97$ is retained only as a descriptive diagnostic.

We then convert this into a deployable, selective-prediction analysis [5]. Non-overlapping 16^3 blocks are ranked by mean entropy and expanded by a 4-voxel halo; we measure net flip utility recovered against a matched-random selector under a paired subject bootstrap (Fig. 3). At the predeclared 20% block budget—executing 29.1% of the foreground-union volume after halo expansion—entropy-ranked regions attain a net utility of 0.161 versus 0.035 for random ($\sim 4.6\times$ in the mean), but under a paired subject bootstrap this advantage is significant only at the smallest 5% budget (lower bound 0.004); at every budget $\geq 10\%$ the paired CI includes zero. With the parameter-matched decoder the deployable region-level opportunity is thus real in the mean but marginal in significance—consistent with a net benefit that is itself barely above zero.

Finally, entropy and confidence are interchangeable routing signals (subject correlations 0.639 vs. 0.648, paired difference ≤ 0.02); Monte-Carlo dropout is inapplicable at this checkpoint’s zero dropout, so we treat confidence as an equally valid baseline and do not claim entropy is uniquely optimal.

2.5. Mechanism: Which Factor Is Removable

The comparisons above use end-to-end models, which share the encoder architecture but not its weights. As a further, causally cleaner check, we *freeze* one trained encoder and train the 2×2 decoder factorial on identical features, then compare Full→channel-only, Full→resolution-only, and Full→combined EffiDec3D [TODO: report per-factor flip rates and boundary profiles from the frozen-encoder run; hypothesis to test: dropping the high-resolution stage yields flips concentrated nearer the boundary, while channel reduction is spatially diffuse and smaller, which would explain why both are removable with little aggregate Dice change]. Because the frozen encoder was optimized for the

full decoder, a near-match by the efficient decoder on those features would be a *conservative* statement of redundancy. We present this regime as validation of the decoder attribution rather than as the primary result.

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